

AI Model Releases and their Use:

1. OpenAI

- **Model:** GPT-5.5
- **Release Date:** April 2026
- **Knowledge Cut-off:** Nov 2025
- **Cost (per 1M tokens):** \$5.00 (Input) / \$30.00 (Output)
- **Context Window:** 1,000,000 tokens
- **Availability:** Public (API/ChatGPT)
- **When to use:** Complex multi-step reasoning, high-stakes decision support, and advanced agentic workflows.

2. Google

- **Model:** Gemini 3.1 Pro
- **Release Date:** May 2026
- **Knowledge Cut-off:** Jan 2026
- **Cost (per 1M tokens):** \$2.00 (Input) / \$12.00 (Output)
- **Context Window:** 2,000,000 tokens
- **Availability:** Public (Google AI Studio/Vertex AI)
- **When to use:** Massively long-context retrieval (e.g., analyzing 20+ PDF documents at once) and multimodal video analysis.

3. Anthropic

- **Model:** Claude 4.7 Opus
- **Release Date:** April 2026
- **Knowledge Cut-off:** Dec 2025
- **Cost (per 1M tokens):** \$5.00 (Input) / \$25.00 (Output)
- **Context Window:** 1,000,000 tokens
- **Availability:** Public (API/Claude.ai)
- **When to use:** Deep creative writing, nuanced ethical reasoning, and complex coding tasks where "human-like" instruction following is critical.

4. Meta

- **Model:** Llama 4 (400B)
- **Release Date:** April 2026
- **Knowledge Cut-off:** Oct 2025
- **Cost:** Free (Open Weights) / API providers vary (~\$2.00/1M tokens)
- **Context Window:** 128,000 tokens
- **Availability:** Public (Open Source)
- **When to use:** Local deployment for data privacy and fine-tuning for specific enterprise niche domains.

5. DeepSeek

- **Model:** DeepSeek-V4-Pro
- **Release Date:** April 2026
- **Knowledge Cut-off:** Nov 2025
- **Cost (per 1M tokens):** \$1.74 (Input) / \$3.48 (Output)
- **Context Window:** 1,000,000 tokens
- **Availability:** Public (API/Open Weights)

- **When to use:** High-performance math, logic, and coding at the lowest possible cost-to-performance ratio.

6. xAI (Elon Musk)

- **Model:** Grok 4.1 Fast
- **Release Date:** March 2026
- **Knowledge Cut-off:** Dec 2025
- **Cost (per 1M tokens):** \$0.20 (Input) / \$0.50 (Output)
- **Context Window:** 2,000,000 tokens
- **Availability:** Public (X Premium+ / API)
- **When to use:** Real-time information processing from social media feeds and large-scale data scraping/summarization.

7. Mistral AI

- **Model:** Mistral Large 3
- **Release Date:** February 2026
- **Knowledge Cut-off:** Nov 2025
- **Cost (per 1M tokens):** \$2.00 (Input) / \$6.00 (Output)
- **Context Window:** 128,000 tokens
- **Availability:** Public (La Plateforme)
- **When to use:** European data sovereignty requirements and high-efficiency multilingual tasks (French, German, Spanish).

8. Alibaba

- **Model:** Qwen 3.6 Plus
- **Release Date:** April 2026
- **Knowledge Cut-off:** Dec 2025
- **Cost (per 1M tokens):** \$0.41 (Input) / \$1.20 (Output)
- **Context Window:** 128,000 tokens
- **Availability:** Public (Alibaba Cloud)
- **When to use:** Southeast Asian language tasks and integrating AI with e-commerce or manufacturing logistics.

9. Cohere

- **Model:** Command R7
- **Release Date:** March 2026
- **Knowledge Cut-off:** Jan 2026
- **Cost (per 1M tokens):** \$1.50 (Input) / \$5.00 (Output)
- **Context Window:** 128,000 tokens
- **Availability:** Public (API/Enterprise)
- **When to use:** Advanced RAG (Retrieval-Augmented Generation) and enterprise tool-use/API orchestration.

10. Databricks

- **Model:** DBRX 2
- **Release Date:** February 2026
- **Knowledge Cut-off:** Oct 2025
- **Cost:** Varies (Usage-based on Databricks clusters)
- **Context Window:** 64,000 tokens

- **Availability:** Public (Open Weights)
- **When to use:** Large-scale SQL generation and data engineering automation within the Databricks ecosystem.

Notes on Transformer:

A **Transformer** is a type of neural network architecture that has become the foundational technology behind modern Artificial Intelligence, including large language models like ChatGPT, Gemini, and Claude. It revolutionized how machines process language by allowing them to understand the **context** and **relationships** between words in a way that was previously impossible.

How It Works: The Core Innovations

Before Transformers, AI processed text one word at a time (sequentially). If a sentence was long, the AI would "forget" the beginning by the time it reached the end. Transformers solved this using three main concepts:

1. *Self-Attention (The "Context" Engine)* - Instead of reading left-to-right, the model looks at every word in a sentence simultaneously.

Example:

- In the sentence "The animal didn't cross the street because it was too tired," the model uses self-attention to realize that "it" refers to the animal.
 - In the sentence "The animal didn't cross the street because it was too wide," the model shifts its attention so "it" refers to the street.
2. *Parallel Processing* - Because Transformers don't have to wait for the previous word to finish processing before starting the next, they can be trained on massive amounts of data using powerful GPUs. This speed is what allowed models to scale from simple chatbots to the massive LLMs.
 3. *Positional Encoding* - Since the model processes all words at once, it technically doesn't know the order of the words. To fix this, it adds a "tag" (positional encoding) to each word to identify its place in the sequence, ensuring it knows the difference between "The dog bit the man" and "The man bit the dog."

Why It Matters - The Transformer architecture shifted AI from being a tool for simple pattern matching to a system capable of:

- **Reasoning:** Understanding complex logic and instructions.
- **Creativity:** Generating coherent essays, poems, and code.
- **Multimodality:** The same architecture is now used to process images (Vision Transformers), audio, and video.

Note: The "T" in **GPT** stands for **Transformer** (Generative Pre-trained Transformer).